WEEK 3 · LESSON 14 OF 24 · HANDOUT

Confusion Matrix, Precision and Recall

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=Kdsp6soqA7o — open the lesson online to watch inside the course.

🎓 **Go deeper:** StatQuest: Sensitivity and Specificity — https://www.youtube.com/watch?v=vP06aMoz4v8

🎓 **Go deeper:** Data School: How to evaluate a classifier in scikit-learn — https://www.youtube.com/watch?v=85dtiMz9tSo

Your goal in this lesson: Build and read a confusion matrix, calculate precision, recall and F1, and decide which matters more for a real problem.

Accuracy is not enough

In Lesson 4 a model that said "nobody defaults" scored 79% accuracy and caught **zero** defaulters. To see what a classifier really does, count its four kinds of answers.

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(11)
n = 2000
income = rng.lognormal(12.2, 0.45, n).round(-3)
loan = rng.uniform(20_000, 300_000, n).round(-3)
months = rng.integers(1, 61, n)
late = rng.poisson(0.6, n)
salary = rng.random(n) < 0.55
z = -3.2 + 2.5 * (loan / income) - 0.03 * months + 0.8 * late - 0.9 * salary
default = (rng.random(n) < 1 / (1 + np.exp(-z))).astype(int)
loans = pd.DataFrame({"income_rwf": income, "loan_rwf": loan, "months_customer": months,
                      "late_payments": late, "salaried": salary.astype(int),
                      "defaulted": default})
print(loans.shape, "default rate:", round(loans["defaulted"].mean(), 3))
loans.head()
```

```
from sklearn.model_selection import train_test_split

X = loans.drop(columns="defaulted")
y = loans["defaulted"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y)
```

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

model = make_pipeline(StandardScaler(), LogisticRegression())
model.fit(X_train, y_train)
```

The confusion matrix

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

pred = model.predict(X_test)
print(confusion_matrix(y_test, pred))
ConfusionMatrixDisplay.from_predictions(y_test, pred, display_labels=["repaid", "default"])
plt.show()
```

You should see:

	Predicted repaid	Predicted default
Really repaid	366 (TN)	28 (FP)
Really defaulted	54 (FN)	52 (TP)

- **True positives (TP) = 52** — defaulters correctly flagged.
- **True negatives (TN) = 366** — good borrowers correctly approved.
- **False positives (FP) = 28** — good borrowers wrongly flagged (a lost customer).
- **False negatives (FN) = 54** — defaulters the model missed (lost money).

"Positive" simply means the class we are looking for (default), not something good.

Precision and recall

> ***Precision*** *= TP ÷ (TP + FP) — of the borrowers we flagged, how many really defaulted?* ***Recall*** *= TP ÷ (TP + FN) — of all real defaulters, how many did we catch?*

```
from sklearn.metrics import precision_score, recall_score, f1_score

print("Precision:", round(precision_score(y_test, pred), 3))   # 52 / 80
print("Recall:   ", round(recall_score(y_test, pred), 3))      # 52 / 106
print("F1:       ", round(f1_score(y_test, pred), 3))
```

- Precision is about **0.65**: when the model says "default", it is right 65% of the time.
- Recall is about **0.49**: it catches only **half** of the real defaulters.

The 84% accuracy hid this weakness completely.

F1: one number for both

> *F1 = 2 × precision × recall ÷ (precision + recall)*

F1 (here about **0.56**) is high only when **both** precision and recall are high. It is useful when the positive class is rare.

The full report

```
from sklearn.metrics import classification_report

print(classification_report(y_test, pred, target_names=["repaid", "default"], digits=3))
```

It shows precision, recall and F1 for each class, plus their averages. **Support** is the number of real examples of each class.

Which matters more?

Situation	Worse mistake	Focus on
Cancer screening	Missing a cancer (FN)	**Recall**
Fraud alerts on mobile money	Missing fraud (FN), but too many alerts annoy customers	Recall, while watching precision
Spam filter	Hiding a real email (FP)	**Precision**
Approving micro-loans	Depends on the money lost vs earned	Both — choose with costs (next lesson)

There is almost always a **trade-off**: catching more positives (higher recall) usually means more false alarms (lower precision). The decision threshold controls it — next lesson.

Sensitivity and specificity

Doctors use other names:

- **Sensitivity** = recall = TP ÷ (TP + FN).
- **Specificity** = TN ÷ (TN + FP) — how many healthy people are correctly cleared. Here 366 ÷ 394 ≈ **0.93**.

Key terms

- **Confusion matrix** — table of TP, TN, FP and FN.
- **Precision** — share of predicted positives that are correct.
- **Recall (sensitivity)** — share of real positives that are found.
- **F1 score** — balance of precision and recall.
- **Specificity** — share of real negatives correctly identified.

Remember this: Always look at the confusion matrix; precision and recall show which mistakes a model makes, and the right balance depends on what each mistake costs.

Name: _____ Date: _____

Activity

Task 1

Compute precision and recall by hand from the confusion matrix, then check with scikit-learn.

Task 2

Build the confusion matrix for the "nobody defaults" `DummyClassifier`. What are its precision and recall?

Task 3

Train a random forest and compare its precision and recall with logistic regression.

Task 4

For a mobile-money fraud detector, write two sentences on whether you would favour precision or recall, and why.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. A model flags 80 borrowers as defaulters and 52 really defaulted. What is its precision?
   A. 0.52
   B. 0.65
   C. 0.49
   D. 0.80
2. There were 106 real defaulters and the model caught 52. What is its recall?
   A. 1.0
   B. About 0.49
   C. 0.65
   D. 0.84
3. In cancer screening, which measure matters most?
   A. Training accuracy
   B. Precision
   C. Recall: catch as many real cancers as possible
   D. Number of features
4. What is a false negative in the loan problem?
   A. A good borrower wrongly flagged
   B. A correct flag
   C. A defaulter the model approved
   D. A correct approval
5. When is F1 high?
   A. Only when precision is high
   B. When accuracy is low
   C. When both precision and recall are high
   D. Only when recall is high
6. For a spam filter, why is precision usually more important?
   A. Spam is harmless
   B. Spam filters have no false positives
   C. A real email hidden as spam is costly
   D. Recall cannot be computed

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